

Probability Distribution

Subject / Theme :

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1. Binomial Distribution (Discrete distribution)
2. Poisson Distribution (Discrete distribution)
3. Normal Distribution (Continuous distribution)

Binomial Distribution:-

A discrete random variable x is said to have binomial distribution if its probability function is as follows

$$f(x) = {}^n C_x p^x \cdot q^{n-x}; \quad x = 0, 1, 2, \dots, n$$

where n and p are the parameters of the distribution and $p + q = 1$.

Here,

n = number of trial

p = probability of success = $\frac{x}{n}$

x = number of success

q = probability of failure.

Assumption of binomial Distribution:-

Four properties of a Binomial Distribution.

1. The sample consists of a fixed number of observations
2. Each observation is classified into one of the two mutually exclusive and collectively exhaustive categories

called success and failure.

3. Probability of an observation being defined as a success p or a failure $1-p$ is constant over all observation.

4. The outcome of any observation is independent of the outcome of any other other observation.

Poisson Distribution:-

A discrete random variable x is said to have poisson distribution if its probability, if its probability is as follows,

$$f(x) = \frac{e^{-m} m^x}{x!}, \quad x = 0, 1, 2, \dots, \infty$$

where, m is the parameter of the distribution and $e = 2.718$.

Practical application of poisson distribution:-

1. The number of suicides reported in a particular day.
2. The number of printing mistakes per page of a book.
3. The number of faculty blades in a packet of 100.
4. The number of letters lost in a mail per day.

Normal Distribution:-

A continuous random variable is said to be have normal distribution if its Probability density function is as follows:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-1/2 \left(\frac{x-\mu}{\sigma} \right)^2}; \quad -\infty \leq x \leq \infty$$

$$-\infty \leq \mu \leq \infty$$

$$\sigma > 0$$

where μ and σ are the parameters of the distribution

Importance of Normal distribution:-

1. In practice under certain condition most of the probability and sampling distributions can be approximated by normal distribution.
2. According to certain limit theorem, if mean and variance of distributions can be converted to normal distribution.
3. Normal distribution is the basis of all sampling distribution.
4. Assumption of normality is the basis of all the test of significance in applied statistics.
5. Normal distributions find its application in industrial statistics such as quality control.

Problem-1

A fair coin is tossed 5 times. Find the probability that,

- i. Exactly two heads.
- ii. No head.
- iii. At least three heads.
- iv. At most two heads.

Sol:-

Here, $n=5$.

$$p=q=\frac{1}{2} \text{ [As the coin is fair]}$$

We know,

$$f(x) = {}^n C_x p^x q^{n-x}; x=0,1,2,\dots,n.$$

$$f(x) = {}^5 C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{5-x}; x=0,1,2,\dots,n$$

Let,

x be the number of head.

i) Here,

$$x=2.$$

$$\therefore f(x) = {}^5 C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{5-2}$$

$$= {}^5 C_2 \cdot \frac{1}{4} \cdot \left(\frac{1}{2}\right)^3$$

$$= \frac{5}{54}$$

$$= 0.0926. (\text{Ans}).$$

ii) Here,

$$x=0.$$

$$\therefore f(0) = {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{5-0}$$

$$= {}^5C_0 \cdot 1 \cdot \left(\frac{1}{2}\right)^5$$

$$= 1 \cdot 1 \cdot \frac{1}{32}$$

$$= \frac{1}{32}$$

(Ans).

iii) At least 3 heads i.e. $x \geq 3$

$$\therefore f(x \geq 3) = f(x=3) + f(x=4) + f(x=5)$$

$$= f(3) + f(4) + f(5)$$

$$= {}^5C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^{5-3} + {}^5C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^{5-4} + {}^5C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^{5-5}$$

$$= 10 \cdot \frac{1}{8} \cdot \frac{1}{4} + 5 \cdot \frac{1}{16} \cdot \frac{1}{2} + 1 \cdot \frac{1}{32} \cdot 1$$

$$= \frac{16}{32}$$

$$= \frac{1}{2}.$$

(Ans).

iv) At most 2 heads, i.e. $x \leq 2$.

$$\begin{aligned} P(x \leq 2) &= P(0) + P(1) + P(2) \\ &= {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{5-0} + {}^5C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{5-1} + {}^5C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{5-2} \\ &= 1 \cdot 1 \cdot \frac{1}{32} + 5 \cdot \frac{1}{2} \cdot \frac{1}{16} + 10 \cdot \frac{1}{4} \cdot \frac{1}{8} \\ &= \frac{1}{32} + \frac{5}{32} + \frac{5}{32} \\ &= \frac{11}{32} \text{ (Ans).} \end{aligned}$$

Problem-2

A family has 5 children. The probability of being a boy or a girl is equal.

Find the probability that has —

- i. No boys.
- ii. Exactly Two boys.
- iii. At most Two boys.
- iv. At least three boys.